

Fall 2026 Exploratory Data Analysis with R

Lecture 6: Hypothesis Testing and Correlation

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01

Recap & Agenda

01 Recap & Agenda

02 Hypothesis Testing

03 Correlation

04 Hypothesis Testing and Correlation with Weights

05 Research Process & The Project

06 Recap

Quick Recap: Last Week

- **Visualization** has a grammar — data, mapping, geometry — and the relationship you're showing points to the right `ggplot2` geometry
- **Dates** are numbers underneath — `lubridate` converts raw strings/numbers into date objects, and `format()` changes only how they're *displayed*
- **Strings** can be extracted, replaced, and split with `stringr`, and regex gives you a compact way to describe a pattern

Today: a new module. We turn from describing a single dataset to asking whether a pattern we see is **real** — a formal way of answering yes/no questions with data.

Goals for Today

- Understand the **logic of hypothesis testing** — null and alternative hypotheses, significance levels, and p -values
- Conduct and interpret **one-sample and two-sample t-tests**, both one-sided and two-sided
- Understand **correlation** — how to measure it, and why it isn't causation
- See how **weights** change the results of both t-tests and correlation
- Connect today's tools to **Project Update #4**

02

Hypothesis Testing

01 Recap & Agenda

02 Hypothesis Testing

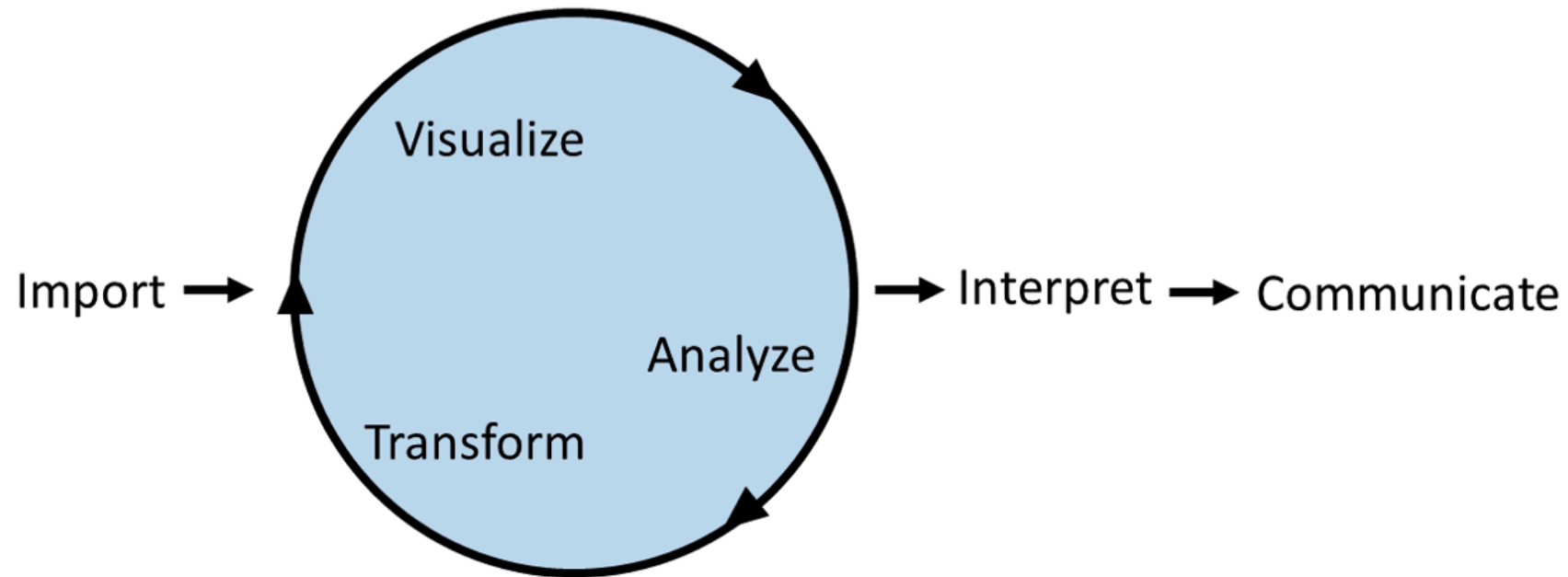
03 Correlation

04 Hypothesis Testing and Correlation with Weights

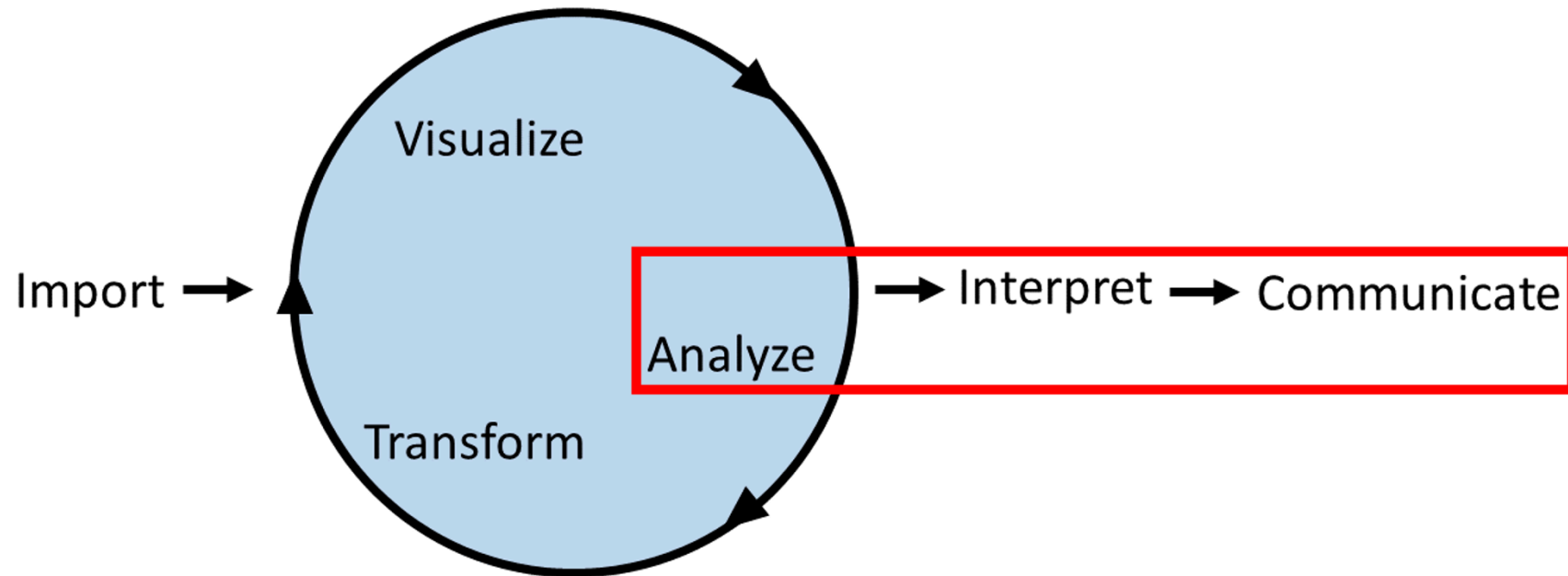
05 Research Process & The Project

06 Recap

How Does Hypothesis Testing Fit into Exploratory Data Analysis?



How Does Hypothesis Testing Fit into Exploratory Data Analysis?



Note: after running a hypothesis test, you might need to go back and transform or visualize the data again — the process is iterative, not a straight line.

The Big Picture

- This lecture is about understanding **relationships in data**.
- **Hypothesis testing** is a formal way of answering yes/no questions using data — for example, questions about a population mean.
- By the end of today, you should be able to conduct and interpret one-sample and two-sample t-tests, including one-sided versus two-sided tests.

Hypothesis Testing Starts From a Question About How the World Works

- Questions that drive hypothesis testing are **specific, yes-or-no** questions, often about the population mean.
- Example: a friend claims that Americans are, on average, 5'7". We might be skeptical and want to assess that claim — hypothesis testing is a way to do this.
- We focus on testing population *means* today, but the same logic applies to a proportion, a standard deviation, or most other population statistics.

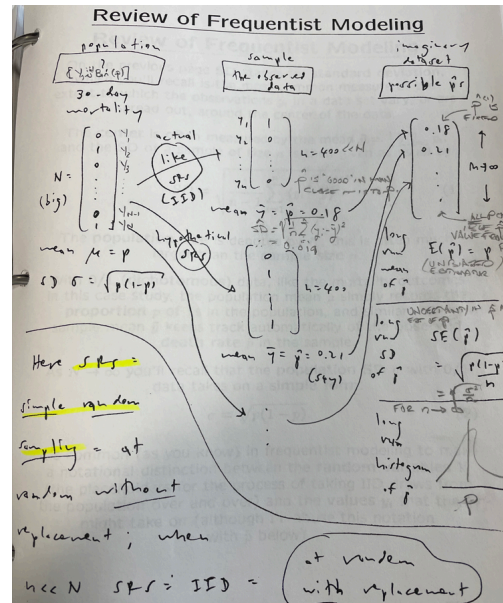
Populations

- To talk about hypotheses, we need to talk about **populations**.
- Examples of populations: all people in the U.S., all households in the world, all firms in the U.S., all banks in the world.
- As researchers, we're trying to learn things about populations.

Populations Are (Generally) Not Observed

- If we observed the height of the *entire* population, we could just check our friend's claim by calculating the population mean.
- We rarely observe the entire population — **we observe a sample**, a subset of the population. Today, we assume the sample is randomly drawn.
- Pizza analogy: if the population is a pizza, the sample is a slice. Hypothesis testing lets us say something about the average pepperoni thickness on the *whole pizza*, having only looked at one slice.

Review of Frequentist Modeling



Formulating Hypotheses

From a question, we formulate two mutually exclusive hypotheses:

- The **null hypothesis** (H_0) — the default outcome, always assumed true. We only *reject* or *fail to reject* it — never *accept* it.
- The **alternative hypothesis** (H_a) — encompasses all other outcomes.
- Our question: is the average American something other than 5'7" in height?
 - Null hypothesis: *Americans are 5'7" on average.*
 - Alternative hypothesis: *Americans are not 5'7" on average.*

What Does a Sample Tell Us About the Population?

We compare our null hypothesis to the sample mean.

- **Population:** U.S. population.
- **Question:** Does the average person in the U.S. have a height that is not 5'7"?
- **Null hypothesis:** The mean height of the U.S. population is 5'7".
- **Alternative hypothesis:** The mean height of the U.S. population is not 5'7".
- **Sample:** A random sample of people in the United States.

Motivation for Hypothesis Testing

- We can't say much about the null hypothesis from just comparing it against the sample mean — the sample's characteristics may not match the population's, and the sample mean might differ from the null just by chance.
- **Hypothesis testing uses statistics** to make an inference about how consistent our sample is with the null hypothesis about the population.

Types of Questions

Questions (and their hypotheses) differ along two dimensions: **how many groups** are being compared, and **what type of comparison** is being made.

- **1 sample** — compare a mean to a fixed point (e.g. *is the average American something other than 5'7"?* — 5'7" is a fixed point, not a second group)
- **2 sample** — compare the means of two groups (e.g. *do men and women have different heights on average?*)
- **2 sided** — the alternative is on both sides of the null (e.g. *not equal to 5'7"* — could be taller **or** shorter)
- **1 sided** — the alternative is on one side of the null (e.g. *taller than 5'4"*)

How Many Groups? What Type of Comparison?

	1 sided	2 sided
1 sample	one group — $H_0: \geq/\leq, H_a: </>$	one group — $H_0: \text{equal}, H_a: \text{not equal}$
2 samples	two groups — $H_0: \geq/\leq, H_a: </>$	two groups — $H_0: \text{equal}, H_a: \text{not equal}$

Types of Questions – Examples

	1 sided	2 sided
1 sample	Is the average American taller than 5'4"?	Is the average American something other than 5'7" in height?
2 samples	Are men taller than women on average?	Do men and women have different heights on average?

Using a Sample to Learn About a Population

After formulating our hypotheses, we choose a **significance level**, α — the probability of rejecting the null hypothesis given that it's actually true (a “type I error”). It's the amount of wrong-call risk we're comfortable with.

- Higher α : more comfortable falsely rejecting a true null.
- Lower α : less comfortable falsely rejecting a true null.
- Traditional levels are 0.10, 0.05, or 0.01 — nothing is special about these, they're just common. R's default is $\alpha = 0.05$.

Next, we run the test and compute the **p -value** — the probability, *given the null is true*, of finding a sample statistic at least as extreme as ours. Then we compare the p -value to α .

Comparing Your p -value to Your α

- If $p \leq \alpha$: finding a sample this extreme would be *rarer* than our tolerance for a false rejection — **we reject the null hypothesis**.
- If $p > \alpha$: finding a sample this extreme isn't rare enough to act on — **we fail to reject the null hypothesis**.
 - **We never accept a hypothesis** — failing to reject is not the same as proving it true.

A Courtroom Analogy

Think of hypothesis testing like a courtroom:

- We ask whether there's enough evidence against the null hypothesis to declare the defendant guilty.
- Rejecting $H_0 \rightarrow$ finding the defendant guilty.
- Failing to reject $H_0 \rightarrow$ finding the defendant *not guilty* — not the same as *innocent*.

Just as a jury cannot declare a defendant innocent, we cannot accept a null hypothesis.

What Increases the Chance of Rejecting a Null Hypothesis?

Before running the test, what features of the data make a lower p -value — and a rejection — more likely?

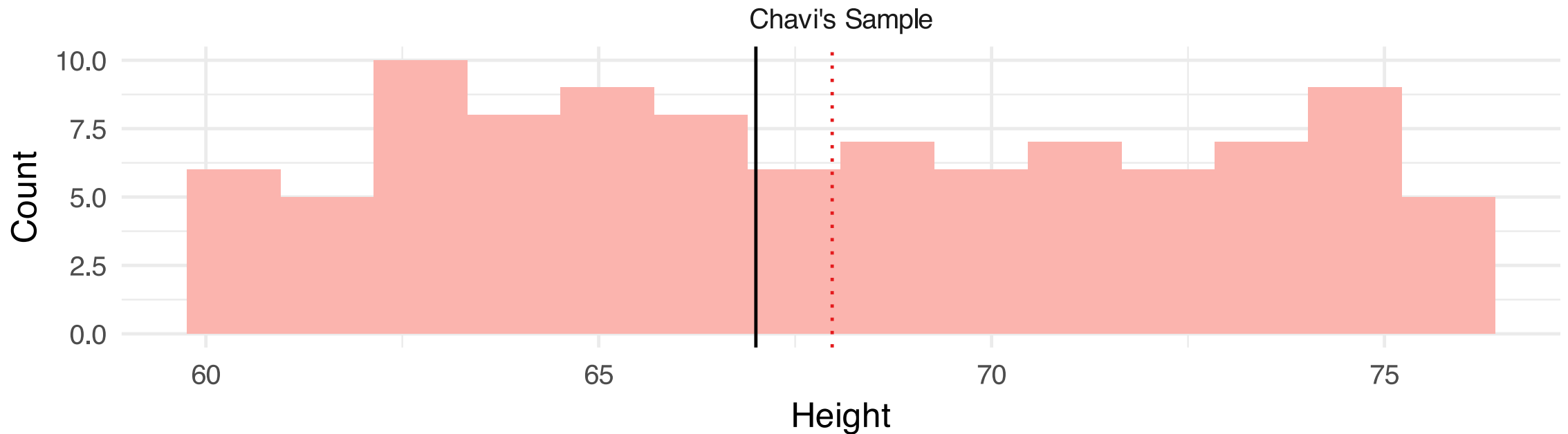
1. **A larger difference** between the sample mean and the hypothesized population mean.
2. **A larger sample size** — larger samples are more representative, so small differences are more likely to be real.
3. **Less variability** — when individuals don't vary much, it's easier to tell a true difference from noise.

Steps of Hypothesis Testing

1. Form the null and alternative hypotheses.
 2. Choose the significance level α .
 3. Run the test to calculate the p -value.
 4. Reject or fail to reject the null hypothesis by comparing the p -value to α .
- **Step 3** in R is almost always a call to `infer::t_test()`, which we'll see in a moment.
 - **Step 4:** if $p \leq \alpha$, we reject the null at the $(1 - \alpha)$ confidence level — a statistically significant difference. If $p > \alpha$, we don't have enough evidence to reject it.

An Example: Four Samples of Height

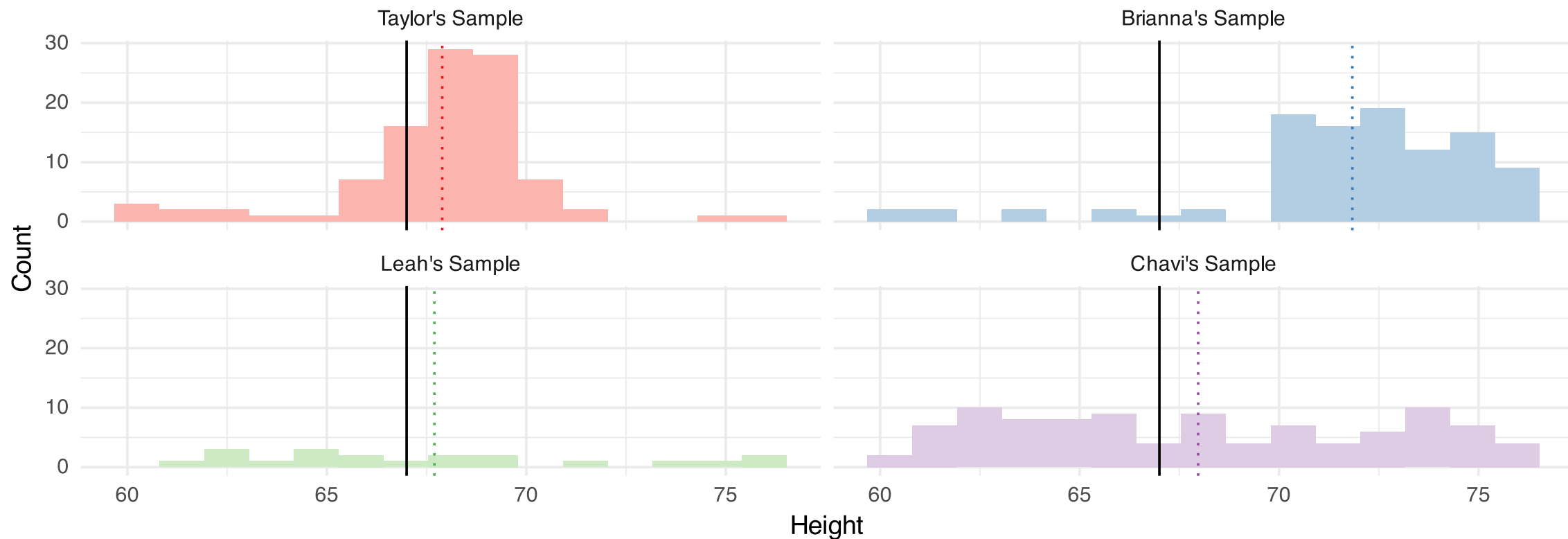
Chavi collects a sample of 100 people's heights to test $H_0 : \mu = 67$ ($H_a : \mu \neq 67$) at $\alpha = 0.05$.



Dotted lines are sample means. Solid line is the null hypothesis at 67".

Her test gives a p -value of 0.041 — she **rejects** the null at $\alpha = 0.05$.

Three Other Students Collect Their Own Samples



Dotted lines are sample means. Solid line is the null hypothesis at 67".

Will their p -values be larger or smaller than Chavi's — and will they reject or fail to reject H_0 ?

A Larger Difference Between the Sample Mean and H_0

Brianna's sample mean sits much farther from 67" than Chavi's.

```
t_test(chavi_data, response = value, mu = 67)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl>
1 2.07 98 0.0410 two.sided 68.0 67.0
68.9
```

```
t_test(brianna_data, response = value, mu = 67)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl>
1 14.4 99 4.56e-26 two.sided 71.8 71.2
72.5
```

A bigger gap between the sample mean and H_0 pulls the p -value down.

A Larger Sample Size

Leah's sample is drawn the same way as Chavi's — just with 20 observations instead of 99.

```
t_test(chavi_data, response = value, mu = 67)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl>
1 2.07 98 0.0410 two.sided 68.0 67.0
68.9
```

```
t_test(leah_data, response = value, mu = 67)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl>
1 0.676 19 0.507 two.sided 67.7 65.5
69.9
```

The same gap is easier to tell apart from noise with a bigger sample — more data, more power.

Less Variability

Taylor's sample is tightly clustered near the null, instead of spread uniformly like Chavi's.

```
t_test(chavi_data, response = value, mu = 67)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr>          <dbl>    <dbl>
1      2.07   98 0.0410 two.sided      68.0    67.0
68.9
```

```
t_test(taylor_data, response = value, mu = 67)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr>          <dbl>    <dbl>
1      3.67   99 0.000394 two.sided      67.9    67.4
68.4
```

Even with a similar sample mean, less variability makes it easier to tell a real difference from noise.

Recap: What Increases the Chance of Rejecting H_0 ?

1. **A larger difference** between the sample mean and the hypothesized population mean.
2. **A larger sample size.**
3. **Less variability.**

One-Sample T-Tests

	1 sided	2 sided
1 sample	one group – $H_0: \geq/\leq, H_a: </>$	one group – H_0 : equal, H_a : not equal
2 samples	two groups – $H_0: \geq/\leq, H_a: </>$	two groups – H_0 : equal, H_a : not equal

FINRA Financial Satisfaction

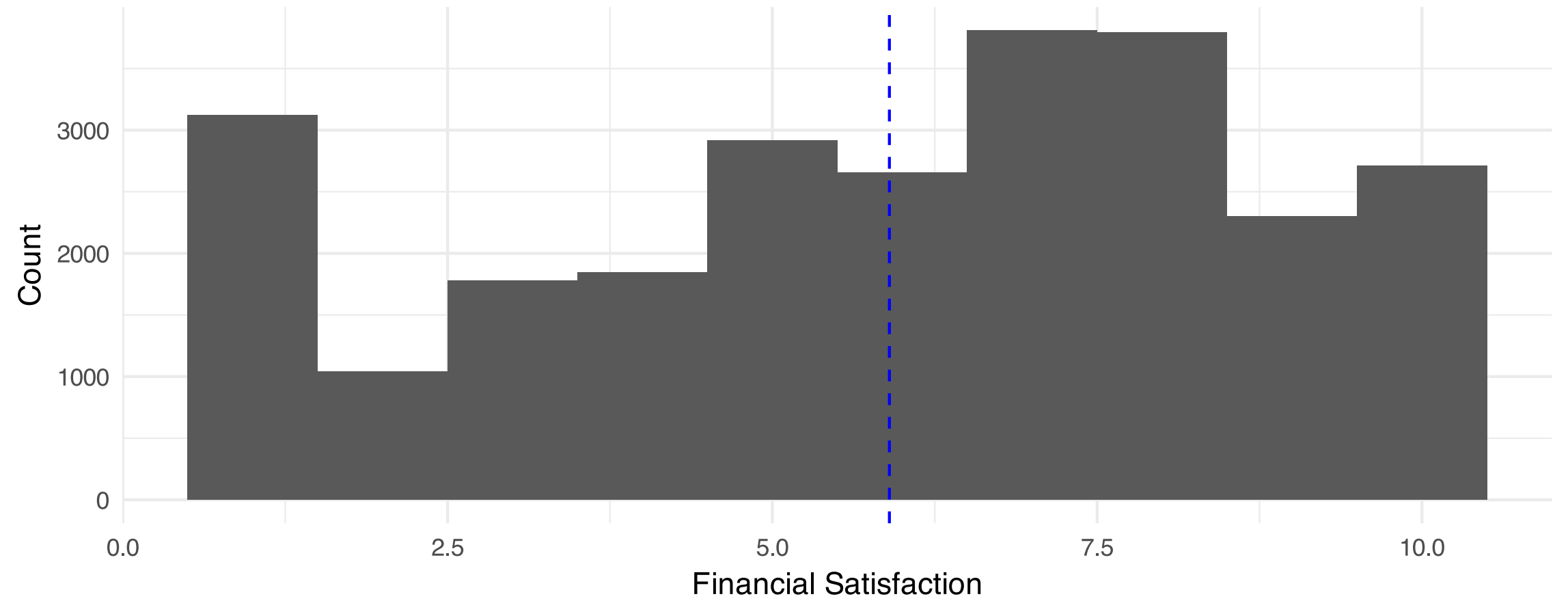
Suppose a study claimed that, across the entire U.S. population, average satisfaction with one's financial state is 5.75 on a scale of 1–10. Let's use the FINRA sample to ask whether *current* satisfaction is different from that claim.

What is the current average level of financial satisfaction, according to our sample?

```
finra_clean %>% pull(fin_satisfied) %>% mean(na.rm = TRUE)
```

```
[1] 5.900608
```

Histogram of `fin_satisfied`



Step 1 & 2: Formulate Hypotheses, Choose α

- $H_0 : \mu = 5.75$, where μ is the mean level of satisfaction with one's personal financial condition
- $H_a : \mu \neq 5.75$
- Let's choose $\alpha = 0.05$.

Step 3: Run the Test

What are the arguments for `t_test`? Check `?infer::t_test`.

```
t_test(data, response = variable, mu = null_mean)
```

- Here, the variable of interest is `fin_satisfied`, and `null_mean` is 5.75 — R computes the sample mean for you automatically.

Step 3: Run the Test

```
t_test(finra_clean, response = fin_satisfied, mu = 5.75)
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>
1      8.66 25967 4.91e-18 two.sided      5.90    5.87    5.93
```

Step 4: Interpret the Results

- Look at the `p_value` column above and compare it to $\alpha = 0.05$: reject, or fail to reject?
- If we reject: **according to our sample, average financial satisfaction is statistically different from 5.75 at the 95% confidence level.**
- The `estimate`, `lower_ci`, and `upper_ci` columns give the sample mean and its confidence interval — the same information `t.test()` prints, just organized as a row instead of a block of text.

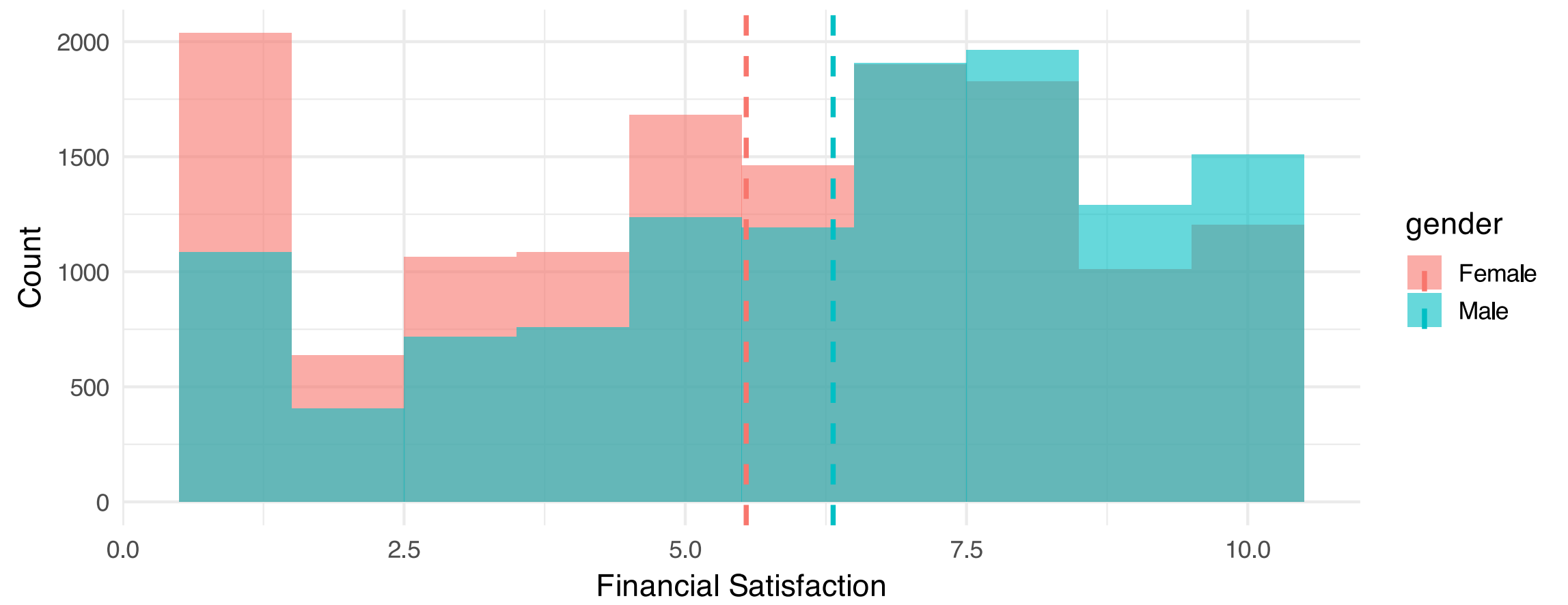
Two-Sample T-Tests

	1 sided	2 sided
1 sample	one group – $H_0: \geq/\leq, H_a: </>$	one group – H_0 : equal, H_a : not equal
2 samples	two groups – $H_0: \geq/\leq, H_a: </>$	two groups – H_0: equal, H_a: not equal

Comparing Means Across Two Groups

- So far we've asked whether the mean of *one* sample differs from a fixed value.
- Now: are the means of *two populations* statistically different from each other? E.g. is average satisfaction with one's personal finances different for men and women?
- We only have sample data for each group — the **two-sample t-test** handles this.

Histograms of `fin_satisfied` by Gender



Step 1 & 2: Hypotheses and Significance Level

- $H_0 : \mu_{women} = \mu_{men}$ — average satisfaction is the same for women and men.
- $H_a : \mu_{women} \neq \mu_{men}$ — average satisfaction differs by gender.
- Is this one-sided or two-sided? We choose $\alpha = 0.05$.

Step 3: Run the Test

In R, a two-sample t-test needs a **response variable** and an **explanatory (group) variable**:

```
t_test(data, formula = response_variable ~ group_variable)
```

- Response variable: `fin_satisfied`
- Group variable: `gender`

```
t_test(finra_clean, formula = fin_satisfied ~ gender)
```

Step 4: Interpret the Results

```
t_test(finra_clean, formula = fin_satisfied ~ gender)
```

Warning: The statistic is based on a difference or ratio; by default, for difference-based statistics, the explanatory variable is subtracted in the order "Female" - "Male", or divided in the order "Female" / "Male" for ratio-based statistics. To specify this order yourself, supply `order = c("Female", "Male")`.

```
# A tibble: 1 × 7
  statistic    t_df    p_value alternative estimate lower_ci upper_ci
  <dbl>    <dbl>    <dbl>    <chr>         <dbl>    <dbl>    <dbl>
1    -22.4 25712. 1.55e-110 two.sided    -0.773   -0.840   -0.705
```

Compare the p -value to $\alpha = 0.05$: is there a statistically significant difference in financial satisfaction between women and men? Notice the warning — `t_test()` always tells you which direction it subtracted in, instead of leaving you to guess.

Two-Sample T-Tests: Three Situations

1. **Two subsets of one variable, split by a group column** — `infer`'s core use case: `t_test(data, response ~ group)`
2. **Two separate variables** (e.g. height recorded in January vs. July) — reshape to one column first with `pivot_longer()`, then it's the same as (1)
3. **Paired differences** (same person, two time points) — `infer`'s formula interface doesn't support `paired`. Fall back to base R: `t.test(x = height_january, y = height_july, paired = TRUE)`

Breakout Session 1

- **Question 1 (one-sample t-test):** Is the average willingness to take financial risks statistically different from 5.0 (out of 10.0)?
- **Question 2 (two-sample t-test):** Is willingness to take financial risks statistically different for those living in the Northeast versus those who do not?
- Hint: the relevant variables are `fin_risks` and `northeast`.

Steps: (1) state H_0 and H_a , (2) choose α , (3) run the test, (4) reject or fail to reject?

Solution: Question 1

- One-sample, two-sided t-test.
- $H_0 : \mu = 5.0$, where μ is willingness to take financial risks. $H_a : \mu \neq 5.0$.
- $\alpha = 0.05$.

```
t_test(finra_clean, response = fin_risks, mu = 5.0)
```

```
# A tibble: 1 × 7
  statistic t_df      p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>      <dbl> <chr>          <dbl>      <dbl>      <dbl>
1      4.50 25967 0.00000695 two.sided          5.08        5.04        5.11
```

Solution: Question 2

- Two-sample, two-sided t-test.
- $H_0 : \mu_{Northeast} = \mu_{non-Northeast}$. $H_a : \mu_{Northeast} \neq \mu_{non-Northeast}$.
- $\alpha = 0.05$.

```
t_test(finra_clean, formula = fin_risks ~ northeast)
```

```
# A tibble: 1 × 7
  statistic t_df      p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>    <dbl> <chr>         <dbl>    <dbl>    <dbl>
1      4.58 6206. 0.00000474 two.sided      0.205    0.117    0.292
```

Two-Sided vs. One-Sided T-Tests

So far we've only used two-sided tests:

- One sample: $H_0 : \mu = \text{number}, H_a : \mu \neq \text{number}$
- Two samples: $H_0 : \mu_{\text{group1}} = \mu_{\text{group2}}, H_a : \mu_{\text{group1}} \neq \mu_{\text{group2}}$

A **one-sided** test can be for one sample ($H_0 : \mu \geq \text{number} / H_a : \mu < \text{number}$, or the mirror image) or two samples ($H_0 : \mu_{\text{group1}} \geq \mu_{\text{group2}} / H_a : \mu_{\text{group1}} < \mu_{\text{group2}}$, or the mirror image).

One-Sided Tests Take a Stand on Direction

- **One-sided tests let you commit to a direction** — you should have a *reason* to only test “greater than” or “less than.” Absent one, default to two-sided.
- Example: you want to know if an intervention **decreases** financial satisfaction.
 - $H_0 : \mu_{intervention} \geq \mu_{no\ intervention}$
 - $H_a : \mu_{intervention} < \mu_{no\ intervention}$

One-Sided Syntax

```
t_test(data, response = variable, mu = null_mean, alternative = "greater")
```

```
t_test(data, response = variable, mu = null_mean, alternative = "less")
```

One-Sided Test of `fin_satisfied`

Suppose we're evaluating whether an intervention **decreased** financial satisfaction. The alternative hypothesis is that the population mean is now *less* than 5.75:

■ $H_0 : \mu \geq 5.75, H_a : \mu < 5.75$

The test rejects the null only if the sample mean is sufficiently *less* than 5.75.

```
t_test(finra_clean, response = fin_satisfied, mu = 5.75, alternative = "less")
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>
1      8.66 25967     1 less           5.90    -Inf     5.93
```

One-Sided Two-Sample T-Test

	1 sided	2 sided
1 sample	one group	one group
2 samples	two groups – $H_0: \geq/\leq, H_a: </>$	two groups – $H_0: \text{equal}, H_a: \text{not equal}$

Does test prep (`testprep`) improve overall score, relative to no test prep (`neither`)?

■ $H_0 : \mu_{testprep} \leq \mu_{neither}, H_a : \mu_{testprep} > \mu_{neither}$

```
t_test(studentdata_clean, formula = totalscore ~ group, alternative = "greater")
```

Warning: The statistic is based on a difference or ratio; by default, for difference-based statistics, the explanatory variable is subtracted in the order "neither" - "testprep", or divided in the order "neither" / "testprep" for ratio-based statistics. To specify this order yourself, supply ``order = c("neither", "testprep")``.

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl>   <dbl> <chr>         <dbl>   <dbl>   <dbl>
1    -4.18  36.9    1.000 greater      -41.2   -57.8    Inf
```

`t_test()` Warns You About Group Order

That p -value came out suspiciously large — and `t_test()` printed a warning explaining why:

The statistic is based on a difference or ratio; by default, for difference-based statistics, the explanatory variable is subtracted in the order "`neither`" - "`testprep`" ... To specify this order yourself, supply `order = c("neither", "testprep")`.

- Without an explicit `order`, `infer` falls back to **alphabetical order** — same as base R — but it *tells you* instead of silently guessing.
- **Just because the code ran doesn't mean it tested the direction you meant.**

Group Order Matters for One-Sided Two-Sample T-Tests

```
t_test(studentdata_clean, formula = totalscore ~ group, alternative = "greater")
```

Warning: The statistic is based on a difference or ratio; by default, for difference-based statistics, the explanatory variable is subtracted in the order "neither" - "testprep", or divided in the order "neither" / "testprep" for ratio-based statistics. To specify this order yourself, supply `order = c("neither", "testprep")`.

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl> <dbl>
1 -4.18 36.9 1.000 greater -41.2 -57.8 Inf
```

The warning tells us the subtraction was `neither - testprep`. We asked for `alternative = "greater"`, so this actually tested $H_a : \mu_{neither} > \mu_{testprep}$ — not what we meant.

Fixing It with **order**

We wanted $H_0 : \mu_{testprep} \leq \mu_{neither}$, $H_a : \mu_{testprep} > \mu_{neither}$ — that is, **testprep** minus **neither**.

- Tell `t_test()` directly which group to subtract first: `order = c("testprep", "neither")`.
- No need to flip "**greater**"/"**less**" in your head to match an alphabetical default — just state the order you mean.

One-Sided Two-Sample T-Test, Correctly Ordered

```
t_test(studentdata_clean, formula = totalscore ~ group,  
       order = c("testprep", "neither"), alternative = "greater")
```

```
# A tibble: 1 × 7  
  statistic t_df p_value alternative estimate lower_ci upper_ci  
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>  
1      4.18  36.9 0.0000852 greater         41.2    24.6    Inf
```

Takeaway: for a one-sided two-sample t-test, always pass an explicit **order** — don't rely on the alphabetical default, even though **infer** warns you about it.

Breakout Session 2

- Is the average student math score greater than 50?
- Hint: the relevant variable is `math_score`.

Steps: (1) state H_0 and H_a , (2) choose α , (3) run the test, (4) reject or fail to reject?

Solution

- One-sample, one-sided t-test.
- $H_0 : \mu \leq 50$, where μ is the average math score. $H_a : \mu > 50$.
- $\alpha = 0.05$.

```
t_test(studentdata_clean, response = math_score, mu = 50, alternative = "greater")
```

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>
1      2.95    59 0.00228 greater         55.7    52.5    Inf
```

Confidence Intervals

`t_test()` always reports a confidence interval alongside the p -value (`lower_ci/upper_ci` columns) — by default a 95% interval (`conf_int = TRUE`), though you can change the level with `conf_level`.

- The confidence level is $1 - \alpha$.
- If the interval **does not contain** the H_0 value, we reject the null — equivalent to $p < \alpha$.
- If the interval **does contain** the H_0 value, we fail to reject — equivalent to $p \geq \alpha$.

Confidence Intervals: Example

```
t_test(finra_clean, response = fin_satisfied, mu = 5.75)
```

```
# A tibble: 1 × 7  
  statistic t_df p_value alternative estimate lower_ci upper_ci  
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>  
1      8.66 25967 4.91e-18 two.sided         5.90     5.87     5.93
```

$\mu = 5.75$ falls *outside* the `lower_ci`–`upper_ci` range above — so we reject H_0 , the same conclusion the p -value gave us.

03

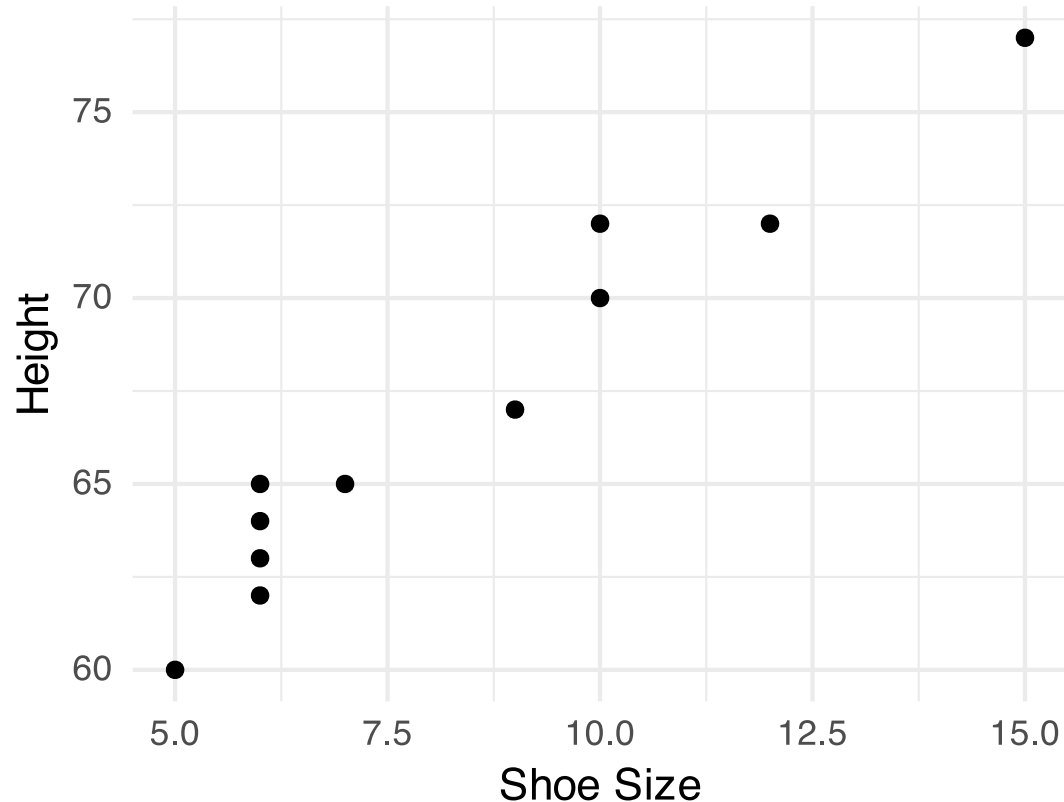
Correlation

- 01 Recap & Agenda
- 02 Hypothesis Testing
- 03 Correlation**
- 04 Hypothesis Testing and Correlation with Weights
- 05 Research Process & The Project
- 06 Recap

What Is Correlation?

- We have two variables of interest, x and y , and we're interested in the relationship between them.
- When x is large, does y tend to be large? Small? Or is there no difference?

Example: Shoe Size and Height



- A large shoe size seems associated with being tall, and a small shoe size with being short — **positively correlated**.
- Does a large shoe size *cause* you to be tall? No.
- Does being short *cause* small feet? No.

Correlation does not imply causation.

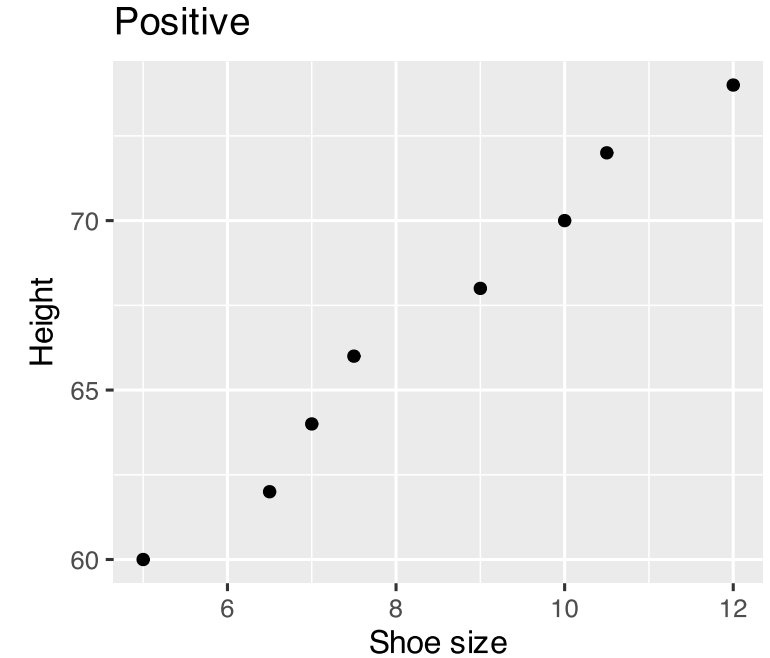
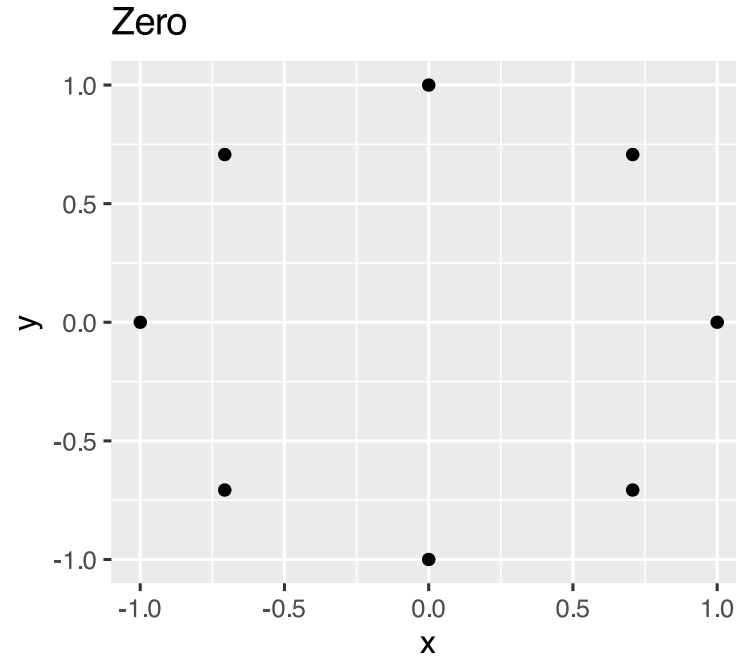
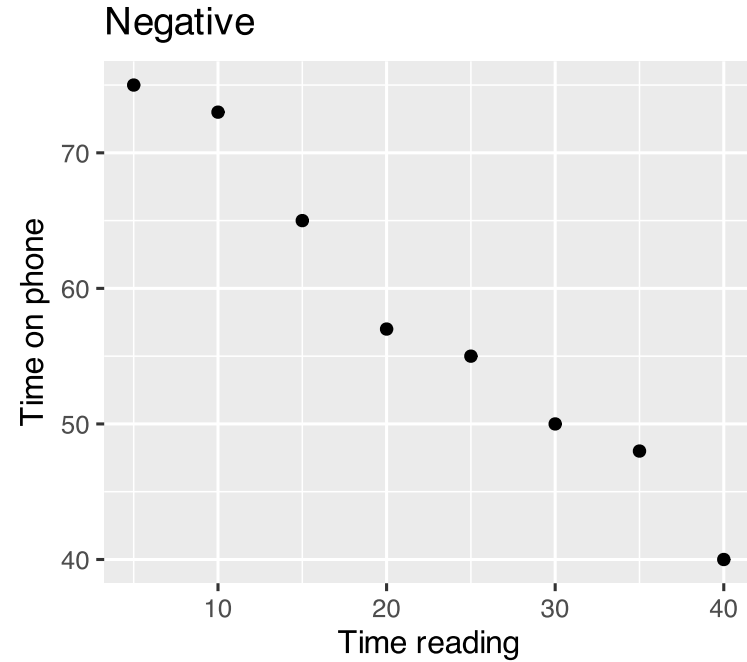
How Do We Measure Correlation?

- Correlation coefficients express both the **sign** and **strength** of a relationship.
- The correlation coefficient of two variables is a number between -1 and 1 , with no units.

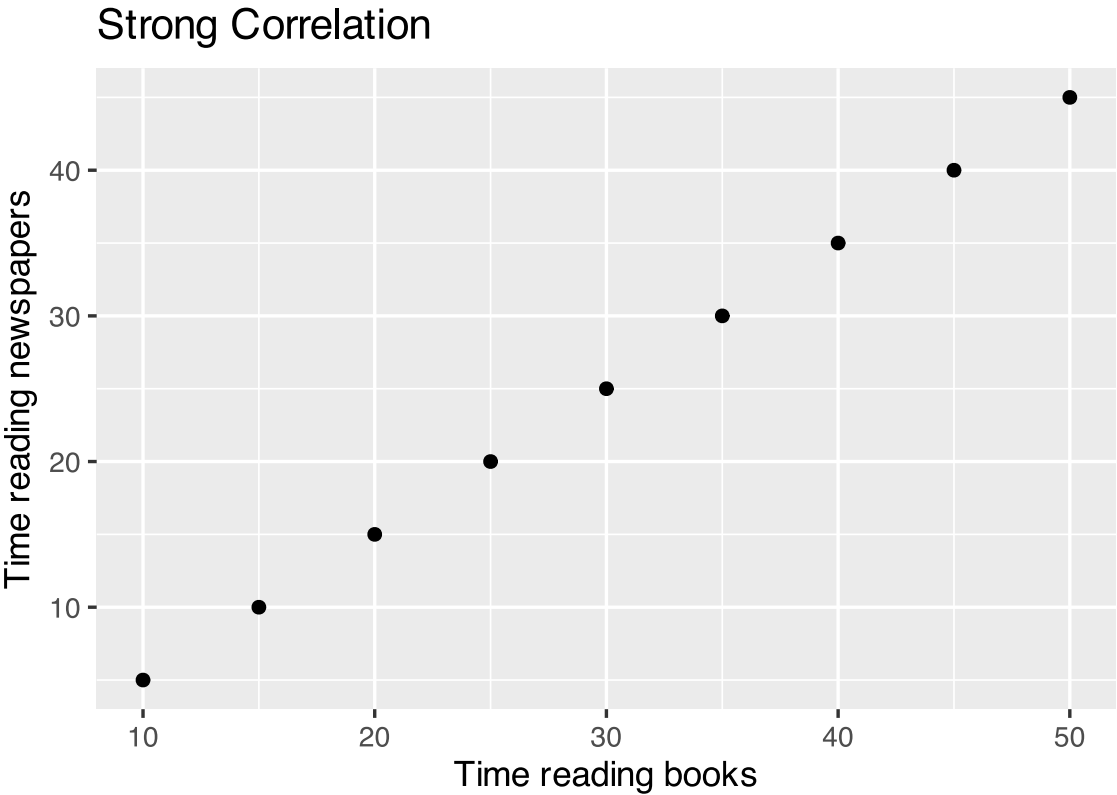
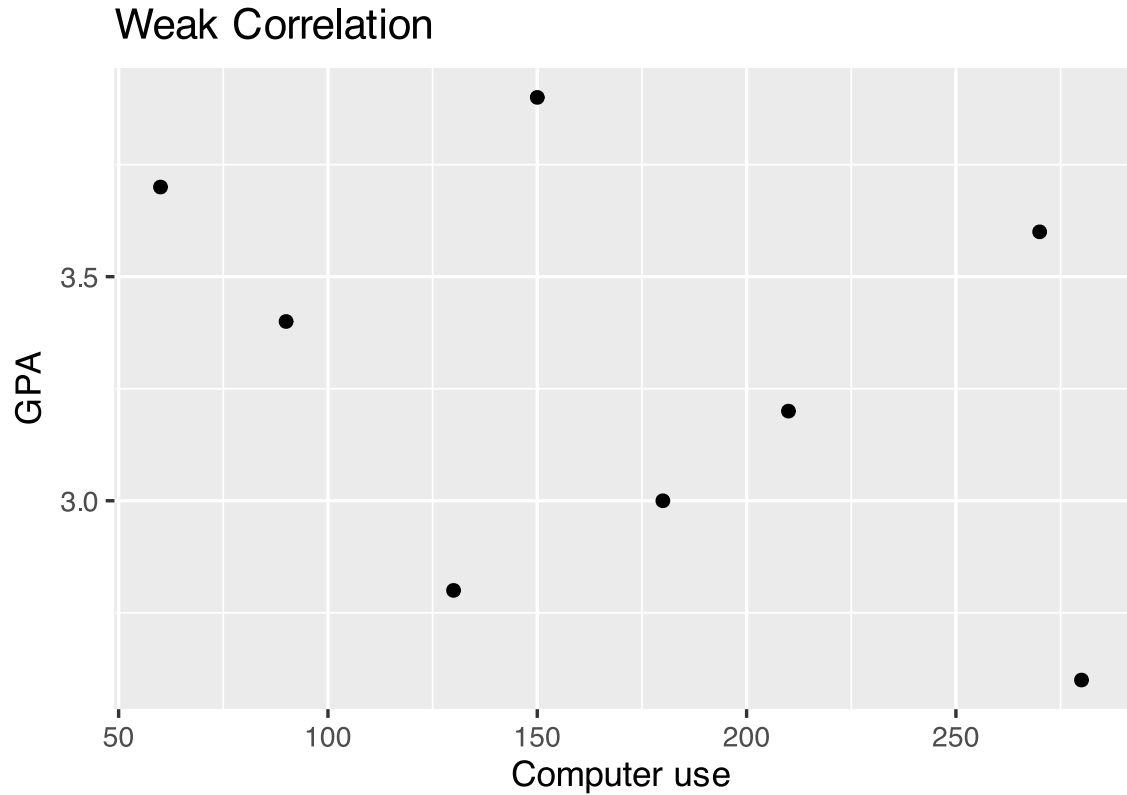
Sign and Strength

- **Sign** — positive or negative. Coefficient in $(0, 1)$: positive relationship. Coefficient in $(-1, 0)$: negative relationship.
- **Strength** — how tight the relationship is. Absolute value near 1: strong. Absolute value near 0: weak.

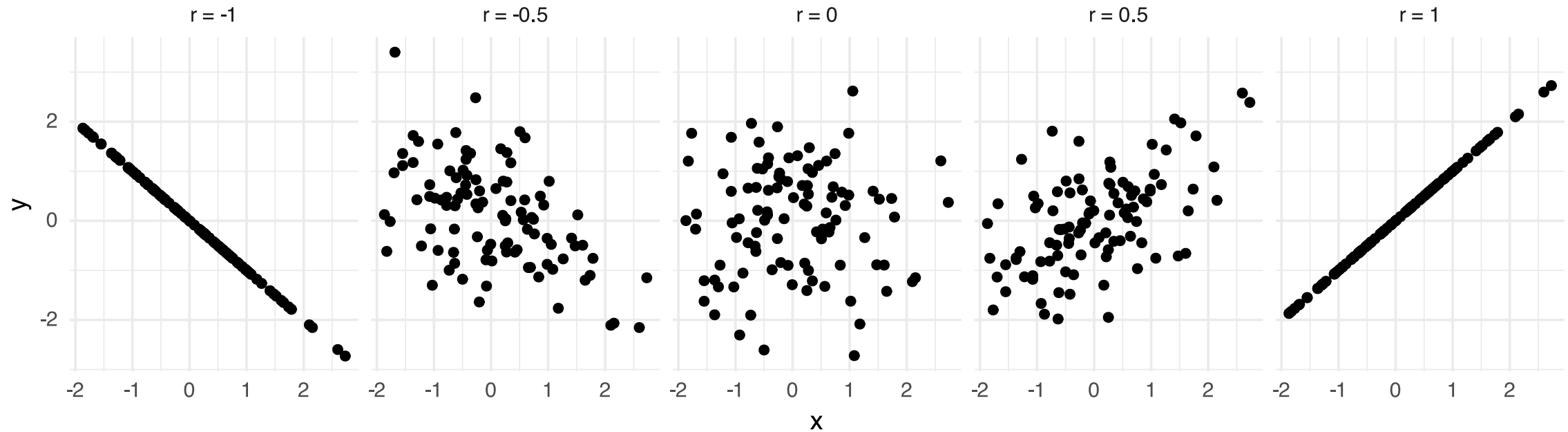
Sign of Correlation



Strength of Correlation



Sign and Strength, Side by Side



Correlation in R

`stats::cor()` computes a correlation coefficient from two numeric vectors:

```
cor(variable1, variable2)
```

How would we compute the correlation of `shoe_size` and `height`?

```
cor(shoe_size, height)
```

Computing Correlation of FINRA Data

We're interested in the relationship between satisfaction with one's personal financial condition (`fin_satisfied`) and willingness to take financial risks (`fin_risks`).

- What sign do we expect? How strong?
- Let's take this question to the data.

Computing Correlation of FINRA Data

```
finra_clean %>%
  summarise(cor = cor(fin_satisfied, fin_risks))
```

```
# A tibble: 1 × 1
  cor
  <dbl>
1 0.307
```

- There's a **positive** relationship between financial satisfaction and willingness to take risks.
- The relationship is not particularly **strong**.

What Else Can `cor` Do?

`cor` can do more than compute a single coefficient — check `?cor`:

```
cor(x, y = NULL, use = "everything",  
    method = c("pearson", "kendall", "spearman"))
```

- For a matrix or data frame passed as `x`, `cor` returns every pairwise correlation at once:

```
finra_clean %>%  
  select(fin_satisfied, fin_risks) %>%  
  cor()
```

Correlation Matrices

`cor` returns a matrix of coefficients when `x` has more than two columns.

1. Create a data frame with only the variables of interest.
2. Compute all correlations at once with `cor()`.

```
data_frame %>%  
  select(col1, col2, col3, ...) %>%  
  cor()
```

Computing a Correlation Matrix of FINRA Data

We're interested in how satisfaction with one's financial condition (`fin_satisfied`), willingness to take risks (`fin_risks`), and feeling of having too much debt (`too_much_debt`) relate to each other. How do you expect them to be related — sign and strength?

```
finra_clean %>%
  select(fin_satisfied, fin_risks, too_much_debt) %>%
  cor() %>%
  knitr::kable()
```

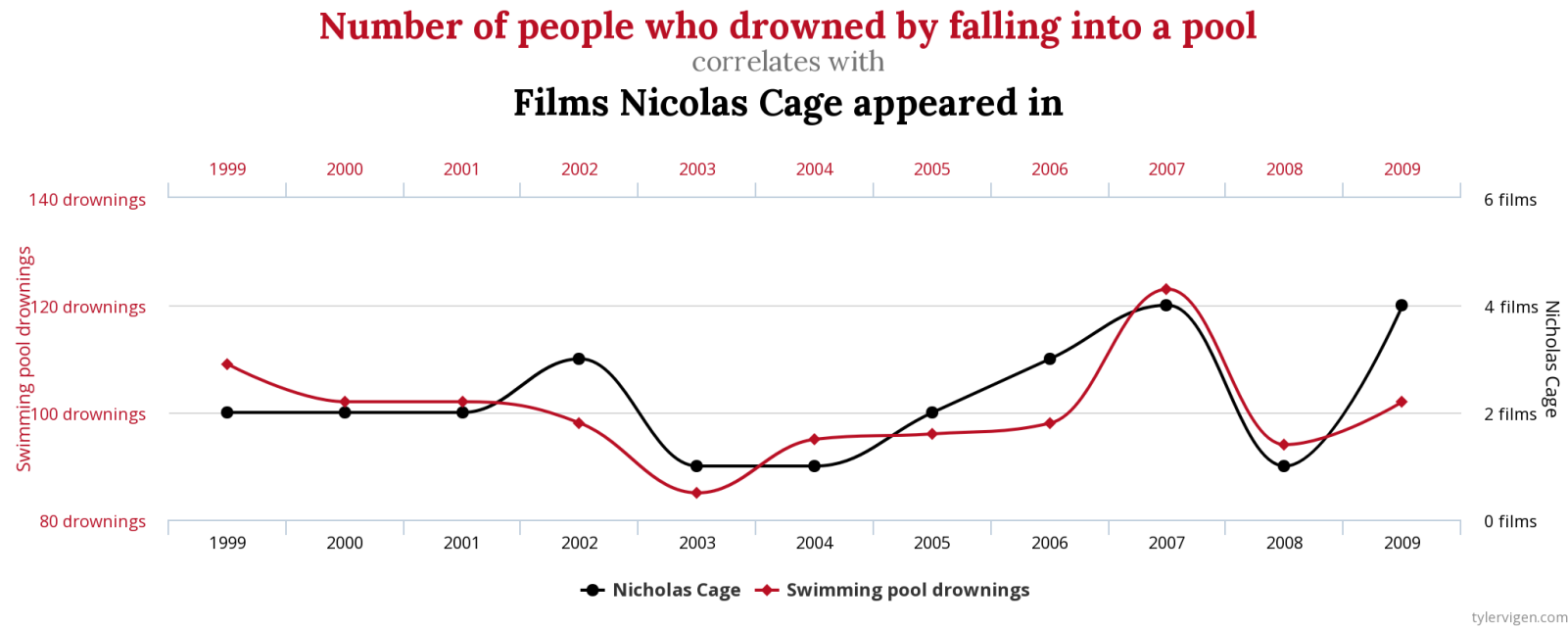
	fin_satisfied	fin_risks	too_much_debt
fin_satisfied	1.0000000	0.3072803	-0.4229149
fin_risks	0.3072803	1.0000000	-0.0014244
too_much_debt	-0.4229149	-0.0014244	1.0000000

Interpreting the Correlation Matrix

- By construction, the 1's on the diagonal tell us every variable is perfectly correlated with itself.
- Financial satisfaction and willingness to take risks are **positively** correlated.
- Financial satisfaction and feeling of having too much debt are **negatively** correlated.
- Willingness to take risks and feeling of having too much debt show **little relationship** at all.

Why bother with a matrix instead of one `cor()` call at a time? It saves time, helps check whether independent variables are highly correlated with each other (worth knowing before a regression), and can flag coding or data-cleaning problems — an unexpectedly extreme correlation (near ± 1) is often a sign something's wrong with how a variable was built.

Correlation and Causation



Source: tylervigen.com/spurious-correlations

- Two variables can be **correlated** without being **causally** related.
- This is called a **spurious relationship**, and can arise by coincidence or a shared time trend.

Key Takeaways About Correlation

1. Correlation coefficients describe the relationship between two variables.
2. A correlation coefficient is unit-less and ranges from -1 to 1 .
3. Negative coefficients imply negative correlation; positive coefficients imply positive correlation.
4. Larger absolute values imply stronger correlation.
5. `stats::cor()` computes correlation coefficients, and correlation matrices.
6. **Correlation does not imply causation.**

04

Hypothesis Testing and Correlation with Weights

- 01 Recap & Agenda
- 02 Hypothesis Testing
- 03 Correlation
- 04 Hypothesis Testing and Correlation with Weights**
- 05 Research Process & The Project
- 06 Recap

Weights

- A sample is an imperfect measure of the population — by chance, the share of a trait in a sample may not match the population's share.
- To account for *known* differences between the sample and the population, we can use **weights**.

A Weighted One-Sample T-Test

Does average satisfaction with one's personal finances equal 5.75?

```
t_test(finra_clean, response = fin_satisfied, mu = 5.75)
with(finra_clean, weights::wtd.t.test(x = fin_satisfied, y = 5.75, weight = national_weight))
```

Unweighted:

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl>
1 8.66 25967 4.91e-18 two.sided 5.90 5.87
5.93
```

Weighted by `national_weight`:

```
$test
[1] "One Sample Weighted T-Test"

$coefficients
      t.value      df      p.value
8.943849 25967.000000 0.000000

$additional
      Difference      Mean Alternative      Std. Err
0.15608693 5.90608693 5.75000000 0.01745188
```

Both reject the same null here, but **weighting changes the sample mean and the p -value.**

`infer::t_test()` has no weighted version, so for the weighted side we still reach for `weights::wtd.t.test()`.

Correlation with Weights

Weights can also change correlation coefficients. `weights::wtd.cors()` computes correlation using weights:

```
wtd.cors(x, y, weight)
```

```
with(finra_clean, weights::wtd.cors(x = fin_satisfied, y = fin_risks, weight =  
national_weight))
```

```
      [,1]  
[1,] 0.2989723
```

Compare that to the unweighted correlation from a couple of slides ago — weighting shifts the result slightly. For your project: if your dataset has weights, you likely want weighted correlation coefficients. Your economist adviser can help you decide.

Weighted T-Test Syntax Reference

- **Two-sided, two-sample** — is the difference in means across two groups significant?

```
wtd.t.test(x = group1_variable, y = group2_variable,
           weight = weight_for_variable1, weighty = weight_for_variable2)
```

- **One-sided, one-sample** — is the mean of a weighted sample different from a hypothesized value?

```
wtd.t.test(x = variable, y = null_mean, weight = weight,
           alternative = "less") # or "greater"
```

- **One-sided, two-sample** — same idea, across two groups. **As with unweighted one-sided tests, check carefully which group R calls group1 vs. group2 before trusting the direction.**

```
wtd.t.test(x = group1_variable, y = group2_variable,  
           weight = weight_for_variable1, weighty = weight_for_variable2,  
           alternative = "less")    # or "greater"
```

05

Research Process & The Project

- 01 Recap & Agenda
- 02 Hypothesis Testing
- 03 Correlation
- 04 Hypothesis Testing and Correlation with Weights
- 05 Research Process & The Project**
- 06 Recap

Research Process: Project Update #4

Project Update #4 forks depending on your data:

- **If your project has weights** — incorporate them into your summary statistics, histograms, and conditional analysis, and compare to your unweighted results. (Today's weighted FINRA examples are exactly this.)
- **If your project doesn't have weights** — incorporate the feedback you've received from your adviser and project buddy so far, and polish your previous visuals and tables.

Either way, you'll also follow up with your adviser and post an update for your classmates.

This Week's Timeline

- **HW #6 — due October 8.** Today's material: hypothesis testing and correlation.
- **Project Update #4 — due October 10.** Weights or feedback, plus the adviser follow-up and classmate post.

Not sure whether your dataset has weights? Ask your adviser before the deadline, not after.

06

Recap

- 01 Recap & Agenda
- 02 Hypothesis Testing
- 03 Correlation
- 04 Hypothesis Testing and Correlation with Weights
- 05 Research Process & The Project
- 06 Recap**

Recap

- **Hypothesis testing** formalizes a yes/no question about a population — form H_0/H_a , pick α , run the test, compare the p -value to α
- Conducted and interpreted **one-sample and two-sample t-tests**, both one- and two-sided — and learned to check how `infer` orders groups
- **Correlation** measures the sign and strength of a relationship between two variables, but never implies causation
- **Weights** change the results of both t-tests and correlation coefficients

Functions Introduced Today

- `infer::t_test()` — one-sample and two-sample t-tests
- `stats::cor()` — correlation coefficients and correlation matrices
- `weights::wtd.t.test()` and `weights::wtd.cors()` — weighted versions of both

Documentation & Further Reading

- `?infer::t_test` — arguments, formula vs. response/explanatory, `order`, `alternative`, `conf_level`
- `?cor` — `method = c("pearson", "kendall", "spearman")`, matrix vs. vector input
- `?weights::wtd.t.test` and `?weights::wtd.cors` — the weighted versions used today

Coming up: loops, debugging, and project discussion.